

Lab 3: Architectural Pattern Selection – Written Justification

Self-Service Coffee Kiosk System

Architecture Selection

We chose **Layered Architecture** for the **Self-Service Coffee Kiosk System**.

Architectural Choice

Layered (N-tier) architecture with three layers. The **Presentation layer** holds the User Interface component (touch screen). The **Business layer** holds the Order Manager, Payment Service and Receipt Printer components. The **Data layer** holds the Database component (menu and pricing). Layers communicate only through interfaces – IOrder, IPayment, IReceipt and IMenuData – and requests flow strictly downward through the Order Manager.

Two Reasons for the Choice

- **The scope is small and the workflow is linear.** The kiosk offers 3 coffees, 2 sizes and card-only payment, and every order follows the same path: select → pay → print receipt. Five components on a single machine do not need independent scaling or deployment, so Microservices would add network calls and operational overhead for no benefit. A Client-Server design would make each kiosk depend on a central server and create a single point of failure. Layered keeps everything in one deployable unit.
- **Layers map directly onto the technical constraints.** Touch-screen handling stays in the Presentation layer, the printer hardware sits behind the Receipt Printer component, and menu/pricing storage sits in the Data layer. Because components depend only on interfaces, a printer, payment provider or menu item can be replaced or added without changing the UI, and the Order Manager keeps all business rules (order flow, price calculation) in one place.

Security Advantage

Strict layering gives a controlled path to sensitive assets. The UI cannot reach the database or payment logic directly; every request passes through the Order Manager, which validates it. Card handling is confined to the Payment Service behind the IPayment interface, so the Order Manager and UI only receive an approved/declined result (never raw card data). This also keeps pricing data safe from tampering through touch input and reduces the part of the system that must be secured for card payments.

Performance Benefit

All layers run on the kiosk itself, so calls between components are fast local calls with no network latency; only the card authorisation travels over the network. The menu and pricing data is tiny (3 coffees × 2 sizes), so the Order Manager can load it once at start-up and serve every touch from memory. The screen therefore responds instantly, and the receipt can be printed right after payment is approved.

Trade-off Acknowledged

Layered designs can add overhead and tight coupling between layers and are hard to scale per component. At this kiosk's size, that overhead is negligible and the simplicity, security and maintainability gains outweigh it.